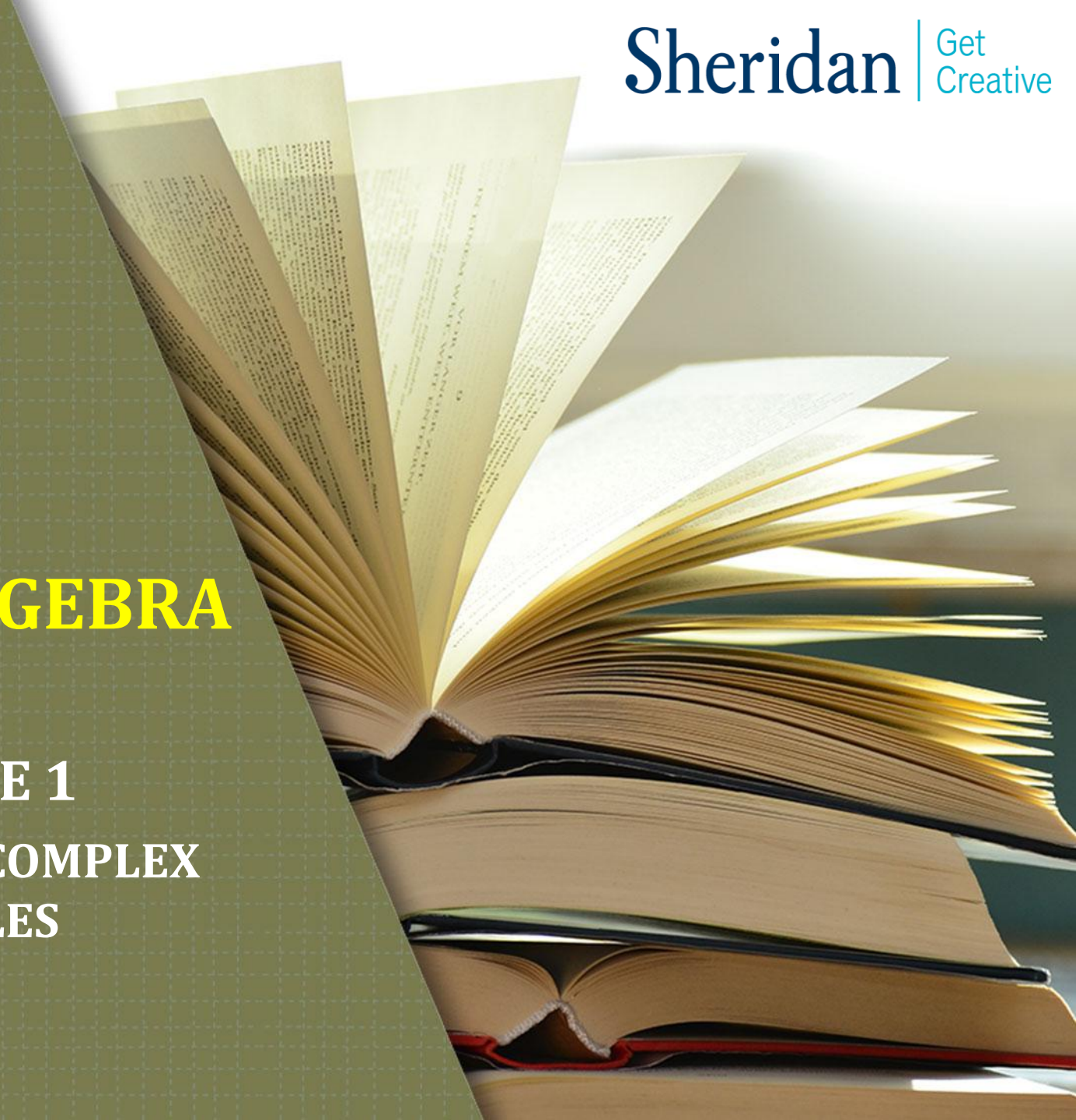


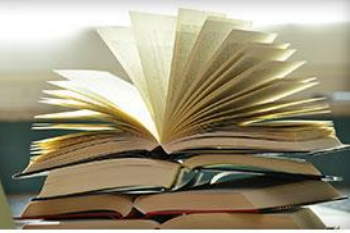
LINEAR ALGEBRA

LECTURE 1

ALGEBRA OF COMPLEX VARIABLES



Complex Numbers In Rectangular Form

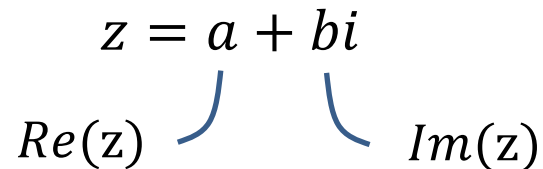


In basic algebra, the variables x and y generally stand for values of real numbers.

The algebra of complex numbers uses the complex variable z to represent a number of the form $a + ib$ or $a + bi$

$$z = a + bi$$

$Re(z)$ $Im(z)$



the “imaginary” number

$$i = \sqrt{-1}$$

Which is assumed to have the property $i^2 = (\sqrt{-1})^2 = -1$

Complex numbers are added, subtracted, and multiplied with the standard rules of algebra but with $i^2 = -1$.

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

$$(a + bi) - (c + di) = (a - c) + (b - d)i$$

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i$$

➤ Exercise 1

Determine the real part and imaginary part of complex numbers.

$$z_1 = 3 + 2i$$

$$z_2 = 1 - 5i$$

➤ Exercise 2

Compute the Multiplication of the complex numbers.

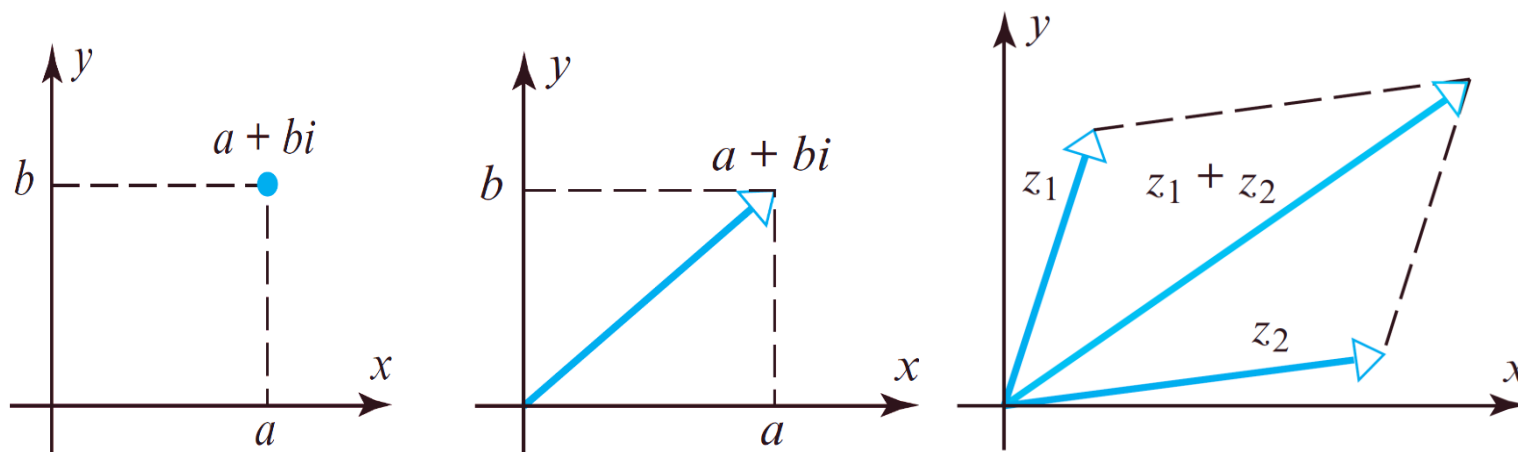
$$(3 - 2i)(4 - 5i)$$

The Complex Plane



Complex numbers can be represented by a point or vector in the xy -plane.

Complex numbers can be added, subtracted, or multiplied by real numbers geometrically.

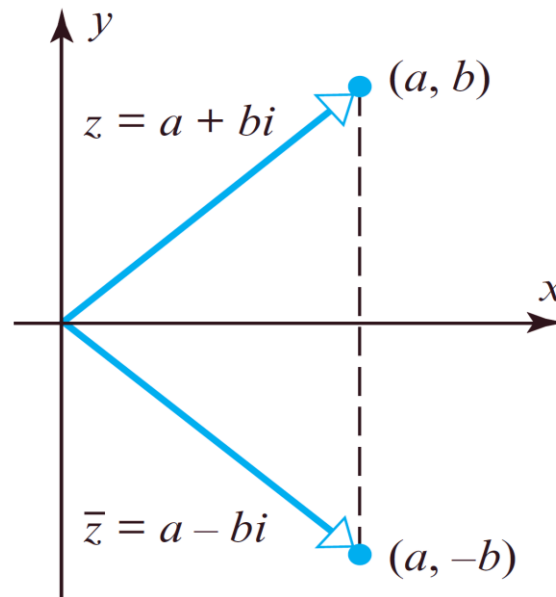


the complex conjugate of z is denoted by $\bar{z}$ and is defined by,

$$\bar{z} = a - bi$$

Numerically, $\bar{z}$ is obtained from z by reversing the sign of the imaginary part.

Geometrically, $\bar{z}$ is obtained by reflecting the vector for z about the real axis.



➤ Exercise 3

Find the complex conjugate of the complex numbers.

$$z_1 = 4 - 5i$$

$$z_2 = 13 + i$$

The following computation shows that the product of a complex number $z = a + ib$ and its conjugate $\bar{z} = a - bi$ is a nonnegative real number,

$$z\bar{z} = (a + ib)(a - ib) = a^2 + b^2$$

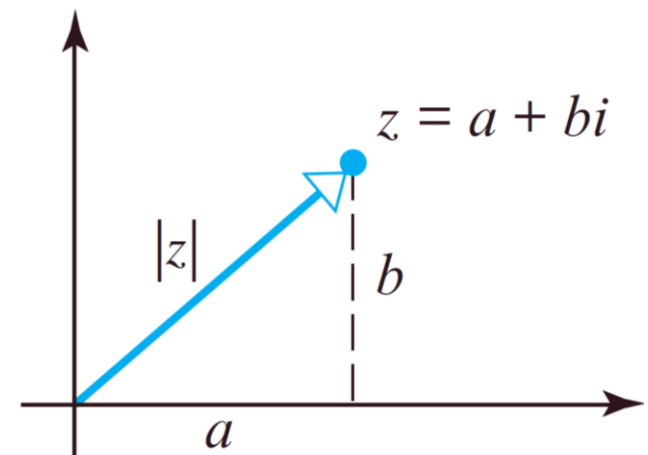
You will recognize that

$$\sqrt{z\bar{z}} = \sqrt{a^2 + b^2}$$

is the length of the vector corresponding to z .

we call this length the modulus (or absolute value of z) and denote it by $|z|$. Thus,

$$|z| = \sqrt{z\bar{z}} = \sqrt{a^2 + b^2}$$





If $z \neq 0$, Then $\left(\frac{1}{z}\right)z = 1$.

This equation has a unique solution for $\frac{1}{z}$, which we can obtain by multiplying both sides by $\bar{z}$ and using the fact that $z\bar{z} = |z|^2$. This yields,

$$\frac{1}{z} = \frac{\bar{z}}{|z|^2}$$

If $z_2 \neq 0$, then the quotient $\frac{z_1}{z_2}$ is defined to be the product of z_1 and $\frac{1}{z_2}$. This yields the formula,

$$\frac{z_1}{z_2} = \frac{\bar{z}_2}{|z_2|^2} z_1 = \frac{z_1 \bar{z}_2}{|z_2|^2}$$

➤ Exercise 4

Find the Modulus of the complex numbers below.

$$z_1 = \sqrt{5} - 5i$$

$$z_2 = 2 + i$$



THEOREM 1 The following results hold for any complex numbers z , z_1 , and z_2 .

1) $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$

2) $\overline{z_1 - z_2} = \overline{z_1} - \overline{z_2}$

3) $\overline{z_1 z_2} = \overline{z_1} \overline{z_2}$

4) $\overline{z_1 / z_2} = \overline{z_1} / \overline{z_2}$

5) $\overline{\overline{z}} = z$

THEOREM 2 The following results hold for any complex numbers z , z_1 , and z_2 .

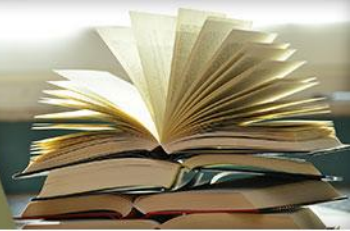
1) $|\overline{z}| = |z|$

2) $|z_1 z_2| = |z_1| |z_2|$

3) $|z_1 / z_2| = |z_1| / |z_2|$

4) $|z_1 + z_2| \leq |z_1| + |z_2|$

Polar Form of a Complex Number



If $z = a + ib$ is a nonzero complex number, and φ is an angle from the real axis to the vector z , then, the real and imaginary parts of z can be expressed as,

$$a = |z|\cos\varphi \quad \text{and} \quad b = |z|\sin\varphi$$

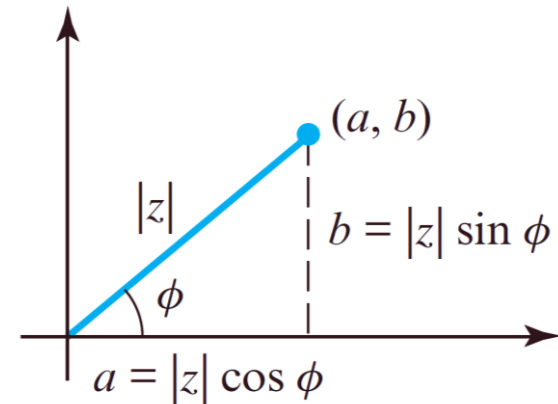
Thus, the complex number $z = a + ib$ can be expressed as,

$$z = |z|(\cos\varphi + i\sin\varphi)$$

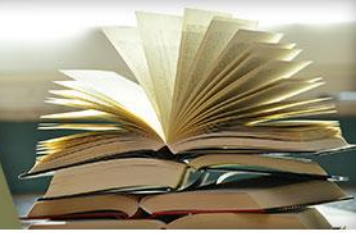
The angle φ in this formula is called an argument of z and is not unique.

Usually, to avoid ambiguity, we choose one special angle called the principal argument, where:

$$-\pi \leq \varphi \leq \pi$$



Geometric Interpretation of Multiplication and Division of Complex Numbers



We will now show how polar forms of complex numbers provide geometric interpretations of multiplication and division. Let,

$$z_1 = |z_1|(\cos\varphi_1 + i\sin\varphi_1)$$

$$z_2 = |z_2|(\cos\varphi_2 + i\sin\varphi_2)$$

Multiplying, we obtain,

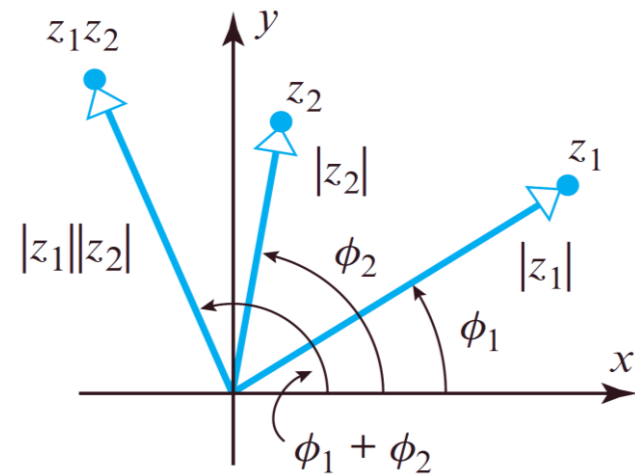
$$z_1 z_2 = |z_1||z_2|[(\cos\varphi_1 \cos\varphi_2 - \sin\varphi_1 \sin\varphi_2) + i(\sin\varphi_1 \cos\varphi_2 + \cos\varphi_1 \sin\varphi_2)]$$

$$z_1 z_2 = |z_1||z_2|[\cos(\varphi_1 + \varphi_2) + i\sin(\varphi_1 + \varphi_2)]$$

That we have applied the trigonometric identities,

$$\cos(\varphi_1 + \varphi_2) = \cos\varphi_1 \cos\varphi_2 - \sin\varphi_1 \sin\varphi_2$$

$$\sin(\varphi_1 + \varphi_2) = \sin\varphi_1 \cos\varphi_2 + \cos\varphi_1 \sin\varphi_2$$



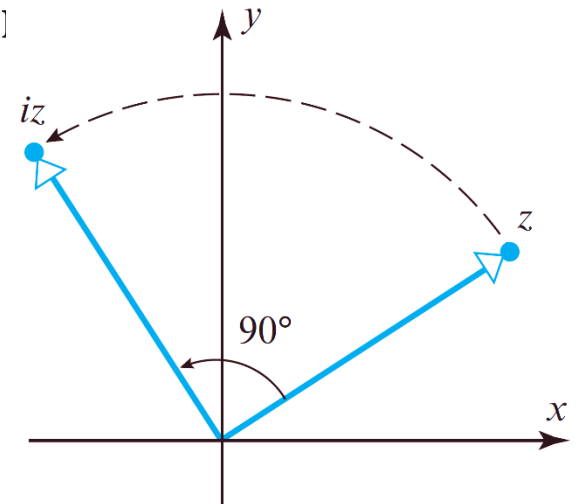
The formula is a polar form of the complex number that has modulus $|z_1||z_2|$ and argument $\varphi_1 + \varphi_2$.

Similar kinds of computations show that,

$$\frac{z_1}{z_2} = \frac{|z_1|}{|z_2|} [\cos(\varphi_1 - \varphi_2) + i\sin(\varphi_1 - \varphi_2)]$$

In other words, angles are added in multiplication, while they are subtracted in division.

Remark The complex number i has a modulus of 1 and a principal argument of $\pi/2$. Thus, if z is a complex number, then iz has the same modulus as z but its argument is greater by $\pi/2$.



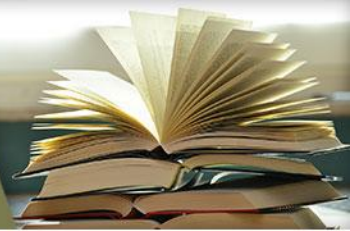
Geometrically, multiplying any complex number by i rotates it 90° counterclockwise without changing its length.

➤ Exercise 5

Use polar forms of the complex numbers $z_1 = 1 + \sqrt{3}i$ and $z_2 = \sqrt{3} + i$ to compute $z_1 z_2$ and z_1 / z_2 .

De Moivre's Formula

duh MWAHV-ruh's



If n is a positive integer, and if z is a nonzero complex number with polar form

$$z = |z|(\cos\varphi + i\sin\varphi)$$

then raising z to the n^{th} power yields,

$$z^n = |z|^n(\cos n\varphi + i\sin n\varphi)$$

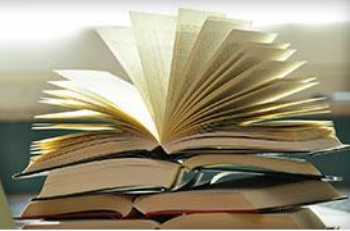
In the special case where $|z| = 1$ this formula simplifies to

$$z^n = \cos n\varphi + i\sin n\varphi$$

which, using the polar form for z , becomes,

$$(\cos\varphi + i\sin\varphi)^n = \cos n\varphi + i\sin n\varphi$$

Euler's Formula



If θ is a real number, say the radian measure of some angle, then the complex exponential function $e^{i\theta}$ is defined to be,

$$e^{i\theta} = \cos\theta + i\sin\theta$$

which is sometimes called Euler's formula.

One motivation for this formula comes from the Maclaurin series in calculus. i.e.

$$\begin{aligned} e^{i\theta} &= 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \frac{(i\theta)^5}{5!} + \frac{(i\theta)^6}{6!} + \dots \\ &= \left(1 - \frac{(\theta)^2}{2!} + \frac{(\theta)^4}{4!} + \dots \right) + i \left(\theta - \frac{(\theta)^3}{3!} + \frac{(\theta)^5}{5!} - \dots \right) \\ &= \cos\theta + i\sin\theta \end{aligned}$$

n^{th} Roots of Complex Numbers



Consider $u = \rho(\cos\alpha + i\sin\alpha)$ is the n^{th} root of complex number z such that, $z = |z|(\cos\theta + i\sin\theta)$ and $u = z^{\frac{1}{n}} = \rho(\cos\alpha + i\sin\alpha)$

Then,

$$|z|(\cos\theta + i\sin\theta) = \rho^n(\cos n\alpha + i\sin n\alpha)$$

So,

$$|z| = \rho^n \quad \text{and} \quad n\alpha = \theta + 2k\pi$$

Thus,

$$\rho = |z|^{\frac{1}{n}} \quad \text{and} \quad \alpha = \frac{\theta}{n} + \frac{2k\pi}{n}$$

Where k is integer from 0 to $n - 1$. So,

$$u = z^{\frac{1}{n}} = |z|^{\frac{1}{n}} \left[\cos \left(\frac{\theta}{n} + \frac{2k\pi}{n} \right) + i \sin \left(\frac{\theta}{n} + \frac{2k\pi}{n} \right) \right]$$

➤ Problem

Solve the equation of $(z + i)^n + (z - i)^n = 0$.

$$\left(\frac{z + i}{z - i}\right)^n = -1 \rightarrow \frac{z + i}{z - i} = \sqrt[n]{-1}$$

So,

$$\frac{z + i}{z - i} = \cos \frac{(2k + 1)\pi}{n} + i \sin \frac{(2k + 1)\pi}{n}$$

$$\frac{z}{i} = \frac{\cos \left(\frac{2k + 1}{n}\right) \pi + 1 + i \sin \left(\frac{2k + 1}{n}\right) \pi}{\cos \left(\frac{2k + 1}{n}\right) \pi - 1 + i \sin \left(\frac{2k + 1}{n}\right) \pi}$$

$$z = i \frac{2\cos^2 \left(\frac{2k + 1}{2n}\right) \pi + 2i \sin \left(\frac{2k + 1}{2n}\right) \pi \cos \left(\frac{2k + 1}{2n}\right) \pi}{-2\sin^2 \left(\frac{2k + 1}{2n}\right) \pi + 2i \cos \left(\frac{2k + 1}{2n}\right) \pi \sin \left(\frac{2k + 1}{2n}\right) \pi}$$

➤ Project: Complex Numbers

If $I = e^{ix_1}e^{ix_2}e^{ix_3} \dots e^{ix_n}$, find the following items.

1) $\sin(x_1 + x_2 + \dots + x_n)$

2) $\cos(x_1 + x_2 + \dots + x_n)$

3) $\tan(x_1 + x_2 + \dots + x_n)$



GOOD LUCK

